

Public Economics (ECON 131)

Section #1: Optimization Review

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1 Economic Motivation

As you learned in Econ 100A/101A, economists generally assume that individuals seek to make choices that will maximize their utility subject to a limited set of resources (most commonly, the individual's total budget or number of hours in a day). In today's section, we will review common optimization techniques to solve problems in economics.

2 Unconstrained Optimization

To solve for a function's maximum/minimum, you will need to first calculate its derivative. Most "well-behaved" functions you will encounter in economics have an interior optimum where the derivative is equal to 0.

1. Write down the function you are optimizing (called the *objective function*; e.g., the utility function).

2. Differentiate the objective function with respect to the choice variable.
3. Set the derivative equal to 0 and rearrange to solve for the choice variable as a function of everything else.

Example: Find the value of x that maximizes $f(x) = \ln(-2x) - 8x^2$.

Solution.

$$\begin{aligned}f'(x) &= \frac{1}{-2x} * (-2) - 16x = 0 \\f'(x) &= \frac{1}{x} - 16x = 0 \\ \frac{1}{x} &= 16x \\ \frac{1}{16} &= x^2 \\ \Rightarrow x &= \pm \frac{1}{4}\end{aligned}$$

Note that only $-\frac{1}{4}$ results in a real-number value of $f(x)$, so we say $-\frac{1}{4}$ is our answer. □

In a more general setting, you will need to check the “second-order” conditions for an interior maximum (perhaps you have actually found a minimum?). In this class, however, we will usually restrict to a class of functions where this will always be satisfied.

In economics, we rarely start off by solving an unconstrained optimization problem. This is because when considering a normal good, the agent always prefers more to less: the unconstrained maximum for a normal good’s utility function would be an infinite amount. Since individuals do not have limitless wealth, though, we are going to usually consider the scenario when individuals have to divide limited resources between at least 2 goods. This class of problems is known as *constrained optimization*.

3 Constrained Optimization

In Econ 131 we will only deal with choices between 2 goods and will generally assume interior solutions. You will not need to worry about second order conditions. However, we will occasionally be interested in whether an individual participates in a scenario. In these cases, you should check whether choosing 0 of one good and spending all wealth on the other good has higher utility than the utility from the interior optimum that you found. We will explicitly remind you when we are working with topics where this is an issue.

The general form of the constrained optimization problems we will consider is:

$$\max_{x,z} U(x,z) \quad \text{s.t.} \quad w = p_x x + p_z z$$

3.1 Mathematical tools: Substitution

When there are only 2 goods to choose between you can re-write the problem as simply choosing how much of one's wealth is spent on one of the goods.

- Once you know how much is spent on good x , you know that the remainder of the person's budget is spent on z (i.e. $(w - p_x x)/p_z$ is spent on z). Thus, in a two good world, the choice boils down to how much to spend on x .
- By using substitution, you can convert the problem into a single variable equation with the constraint built in, and now you can solve this the same way you solved the unconstrained maximization.

We then have:

$$w = p_x x + p_z z \Rightarrow z = \frac{w - p_x x}{p_z}$$
$$\max_{x,z} U(x, z) \text{ s.t. } w = p_x x + p_z z \Leftrightarrow \max_x U\left(x, \frac{w - p_x x}{p_z}\right)$$

1. Now take the total derivative of our re-written utility function with respect to x and set it equal to 0.¹
2. Solve for x to find the optimal x .
3. Substitute x into $z = \frac{w - p_x x}{p_z}$ to find the optimal z .

3.2 Mathematical tools: Lagrangian optimization

An alternative method for optimization is the Lagrangian method, which allows us to embed the budget constraint in the maximization. Using the same utility function and budget constraint as above, we can express the problem as follows:

$$\max_{x,z,\lambda} \mathcal{L}(x, z, \lambda) = \max_{x,z,\lambda} U(x, z) + \lambda[w - p_x x - p_z z], \quad (1)$$

where λ is a new variable which we have added to the problem, called the *Lagrange multiplier* on the constraint. A couple notes:

- We have set up $\mathcal{L}$ such that the term in brackets *should* always be weakly positive (you can not spend more than your budget). Suppose λ is positive: if you were to spend more than your budget, then your utility would go down. For this reason, we often think of λ as a "penalizer" term for violating the constraint.
- When the constraint is fulfilled and the budget is met, then the expression in brackets becomes 0 (including the multiplier). In other words, at the optimum, it must be the case that the maximum for $\mathcal{L}$ and the maximum for U coincide – so we can solve everything in terms of

¹Note that in a general setting without an explicit functional form for $U(\cdot, \cdot)$, you will need to evaluate a *total derivative*. For this class, though, you will likely get an explicit form for U .

$\mathcal{L}$! Another way to think about this is that $\mathcal{L}$ incorporates the costs of choosing more of x – namely, you can not get as much z – via inclusion of the budget constraint.

Note that $\mathcal{L}$ is now expressed in terms of an unconstrained problem, and so we can leverage the standard machinery from before. Let us take the first order conditions for maximization of $\mathcal{L}$:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial U}{\partial x} - \lambda p_x = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial z} = \frac{\partial U}{\partial z} - \lambda p_z = 0 \quad (3)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = w - p_x x - p_z z = 0 \quad (4)$$

The first term on the RHS of (2) is usually called the *marginal utility* in x :

$$MU_x := \frac{\partial U}{\partial x}$$

Cancelling out λ from (2) and (3), we obtain:

$$\frac{MU_x}{p_x} = \frac{MU_z}{p_z} \quad (5)$$

Let us interpret: at an optimum, the marginal utility per dollar spent on X should be equal to the marginal utility of a dollar spent on Z . This idea is usually called the **equimarginal principle**. Let us rearrange to obtain an equivalent (and graphical) interpretation of the first order conditions:

$$MRS_{xz} := \frac{MU_x}{MU_z} = \frac{p_x}{p_z} \quad (6)$$

This expression should be familiar to you from Econ 100A/101A: the marginal rate of substitution (the rate at which a consumer is willing to trade off between two goods) must be equal to the price ratio at any *interior* optimum.²

Knowing these formulas for interior optima can be a useful shortcut for solving optimization problems!

²There are cases in which this will not work, such as: non-differentiable utility functions (e.g. perfect complements), non-strictly concave indifference curves (e.g. perfect substitutes), and negative tangency. You should be able to recognize the first two immediately, and we will need to check to make sure that we are always finding demand to be positive.

3.3 Example 1: Homer Simpson, Pizza, and Beer

Homer Simpson has \$20 to spend on pizza slices (s) and beer bottles (b). A pizza slice costs \$1, while beer costs \$2. Homer's utility is $U(s, b) = s^{1/2}b^{1/2}$. How many pizza slices and beers should Homer choose to maximize his utility? (Try solving this using substitution, the Lagrangian, and the equimarginal principle).

1. **Substitution.** The budget constraint is:

$$s + 2b = 20 \Leftrightarrow s = 20 - 2b$$

Substitute into the utility function:

$$U(b) = (20 - 2b)^{1/2}b^{1/2}$$

The first order conditions (with respect to our single choice variable b):

$$\frac{1}{2}(20 - 2b)^{-1/2}(-2)b^{1/2} + \frac{1}{2}b^{-1/2}(20 - 2b)^{1/2} = 0 \quad (7)$$

$$\Rightarrow \frac{b^{1/2}}{(20 - 2b)^{1/2}} = \frac{(20 - 2b)^{1/2}}{2b^{1/2}} \quad (8)$$

Solving for b :

$$(20 - 2b) = 2b \Rightarrow 20 = 4b \Rightarrow b = 5$$

Now substitute $b = 5$ into the budget constraint to solve for s :

$$s = 20 - 2 * 5 \Rightarrow s = 10 \Rightarrow b = 5$$

Homer buys 10 slices of pizza and 5 beers

2. **Using the Lagrangian Method.** The Lagrangian is:

$$\mathcal{L}(s, b, \lambda) = s^{1/2}b^{1/2} + \lambda(20 - s - 2b)$$

Take the FOCs:

$$\frac{\partial \mathcal{L}}{\partial s} = \frac{b^{1/2}}{2s^{1/2}} - \lambda = 0 \quad (9)$$

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{s^{1/2}}{2b^{1/2}} - 2\lambda = 0 \quad (10)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 20 - s - 2b = 0 \quad (11)$$

Cancel out λ from (9) and (10) - this should be familiar!

$$\frac{s^{1/2}}{2b^{1/2}} = 2 \times \frac{b^{1/2}}{2s^{1/2}} \Rightarrow s = 2b$$

Substitute into the budget constraint:

$$20 - 2b - 2b = 0 \Rightarrow b = 5$$

$$s = 2b \Rightarrow s = 10$$

Homer buys 10 slices of pizza and 5 beers

3. **Equimarginal Principle.** Begin by taking FOCs of choice variables:

$$\frac{\partial U}{\partial s} = \frac{b^{1/2}}{2s^{1/2}}, \quad \frac{\partial U}{\partial b} = \frac{s^{1/2}}{2b^{1/2}}, \quad p_s = 1, \quad p_b = 2$$

Now set the ratio of FOCs equal to the ratio of prices:

$$\frac{\partial U / \partial s}{\partial U / \partial b} = \frac{P_s}{P_b} \Rightarrow \left(\frac{b^{1/2}}{2s^{1/2}} \right) \left(\frac{2b^{1/2}}{2s^{1/2}} \right) = \frac{1}{2}$$

Solve:

$$\frac{2b}{2s} = \frac{1}{2} \Rightarrow \frac{b}{s} = \frac{1}{2}$$

Homer buys 10 slices of pizza and 5 beers

Note: Homer chooses to spend \$10 on beer and \$10 on pizza - i.e. he spends half of his budget on each. Notice that both s and b are raised to the $\frac{1}{2}$ in Homer's utility function. *This is not a coincidence.*

4 Cobb-Douglas Preferences

Utility functions of the form $U(A, B) = A^\alpha B^\beta$ with $\alpha + \beta = 1$ are called "Cobb-Douglas" preferences. Functions of this form are used all the time in economics because they are very user friendly and have some nice properties.

- **General Form:** $U(A, B) = A^\alpha B^\beta$ with $\alpha + \beta = 1$
- **Example:** $U(A, B) = A^{0.25} B^{0.75}$

4.1 Useful properties of C-D preferences

1. They can be re-written as $U(A, B) = A^\alpha B^{1-\alpha}$
2. They represent two normal goods with positive diminishing marginal utility in each good.

$$\begin{aligned}\frac{\partial U}{\partial A} &= \alpha A^{\alpha-1} B^{1-\alpha} & \frac{\partial^2 U}{\partial A^2} &= \alpha(\alpha-1) A^{\alpha-2} B^{1-\alpha} \\ \frac{\partial U}{\partial B} &= (1-\alpha) A^\alpha B^{-\alpha} & \frac{\partial^2 U}{\partial B^2} &= -\alpha(1-\alpha) A^\alpha B^{-\alpha-1}\end{aligned}$$

3. The optimal allocation of resources across the two goods will have a predictable form.
4. General form: Let p_A = the price of each unit of A, and p_B = the price of each unit of B. Then solving

$$\max_{A,B} U(A, B) = A^\alpha B^{1-\alpha} \quad \text{s.t.} \quad w = p_A A + p_B B$$

results in

$$A^* = \frac{\alpha w}{p_A}, \quad B^* = \frac{(1-\alpha)w}{p_B}$$

5. One last trick: for any general utility function $U(A, B) = A^\alpha B^\beta$, the optimal A^*, B^* also solve

$$\log U(A, B) = \alpha \log A + \beta \log B$$

which is a very nice function to differentiate. This trick holds for any **monotone** (increasing or decreasing) transformation f of the original utility function.

4.2 Economic Implications

1. *This will save you time on exams!*
2. This implies that individuals with these types of preferences devote a constant share of wealth to each good. The share of wealth devoted to each good corresponds to the exponent on that good in the utility function. In the above example, the individual spends αw on A and $(1-\alpha)w$ on B.
3. Notice, the price of B does NOT appear in the formula for the optimal amount of A. Similarly the price of A appears nowhere in the optimal amount of B.

4.3 Example 1: Homer's problem, revisited

1. Knowing what we know now, how could we have solved Homer's problem in a faster way?
 - Homer's utility function is: $U(s, b) = s^{1/2}b^{1/2}$.
 - First: check that Homer's utility function is Cobb-Douglas: $\frac{1}{2} + \frac{1}{2} = 1$
 - Therefore we know that he spends $\frac{1}{2}$ of his budget on pizza and $\frac{1}{2}$ on beer. Therefore he spends \$10 on each.
 - If pizza slices cost \$1, then \$10 will buy **10 slices of pizza**.
 - If beer costs \$2, then \$10 will buy **5 beers**.
 - Thus we get the same answer as before with *much less work*!

Let us try out the trick from above. Consider the alternate utility function representation:

$$\tilde{U}(s, b) = \log(U(s, b)) = \frac{1}{2} \log s + \frac{1}{2} \log b$$

2. Is this a monotonic transformation of the original utility function? Why or why not?
Solution: Yes, because the logarithmic function is strictly increasing.
3. Find Homer's MRS using this representation. How does this relate to the MRS we found before? What does this imply for how much beer and pizza Homer chooses?

Solution:

$$\begin{aligned}\frac{\partial U}{\partial s} &= \frac{1}{2} \frac{1}{s}, & \frac{\partial U}{\partial b} &= \frac{1}{2} \frac{1}{b}, \\ \frac{\partial U / \partial s}{\partial U / \partial b} &= \frac{b}{s}\end{aligned}$$

which is exactly the MRS we found in the previous problem. This implies that the problem has the same solution as in the previous problem.

4.4 Example 2: Sam and labor supply

Sam is a substitute teacher. Every week Sam has to choose how many hours to work (L) and how many to spend on other activities (leisure). Sam earns \$10 for every hour worked and therefore has $\$10 \cdot L$ to spend on all other goods. (There are 168 hours in a week.) The price of all other goods has been normalized to 1.³ Sam's utility is: $U(c, h) = \frac{1}{6}ch - (h - 20)^2$, where h is the number of hours of leisure Sam has.

1. Solve for c and h using $\frac{\partial U / \partial c}{\partial U / \partial h} = \frac{P_c}{P_h}$.

- First, note that p_h is \$10 (the wage Sam would have earned had they worked that hour).
- We can also write the budget constraint as: $c = 10L \Rightarrow c = 10(168 - h) = 1680 - 10h$.
- Then we can write

$$\frac{\partial U}{\partial c} = \frac{1}{6}h, \quad \frac{\partial U}{\partial h} = \frac{1}{6}c - 2(h - 20)$$

$$\frac{1/6h}{1/6c - 2(h - 20)} = \frac{1}{10} \Rightarrow 10/6h = 1/6c - 2(h - 20) \Rightarrow 22h = c + 240$$

- Now plug in for c using the budget constraint:

$$22h = -10h + 1680 + 240$$

$$h = 60$$

- **Sam chooses 60 hours of leisure. Thus, Sam works 108 hours and earns \$1,080.**

2. Now, Texas introduces a 20% wage tax. How much does Sam choose to work now?

- Note that the only thing that has changed in $\frac{\partial U / \partial c}{\partial U / \partial h} = \frac{P_c}{P_h}$ is the price ratio: now $\frac{P_c}{P_h} = \frac{1}{8}$.
- Now, choosing to work for one hour less only involves giving up \$8 of income.
- The marginal rates of substitution don't change, so they do not need to be re-calculated.

$$\frac{1/6h}{1/6c - 2(h - 20)} = \frac{1}{8} \Rightarrow 8/6h = 1/6c - 2h + 40 \Rightarrow 20h = c + 240$$

- Plug in for c using the budget constraint, $c = 8L = 8(168 - h)$:

$$20h = 8(168 - h) + 240$$

$$h \approx 56.6$$

- **Sam chooses 56.6 hours of leisure. Thus, Sam works 111.4 hours and earns \$891.**

³Self-test: is this a restrictive assumption?